

Portable Audit eXporter (PAX) - Purview Audit Log Exporter

 **Quick Start:** Download the script → [PAX_Purview_Audit_Log_Processor_v1.6.0.ps1](#)

Script: [PAX_Purview_Audit_Log_Processor_v1.6.0.ps1](#)

Version: 1.6.0

Audience: IT admins, security/compliance analysts, BI/data teams

Runtime: PowerShell 5.1 (compatible) / PowerShell 7+ (recommended)

License: MIT

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Overview

What It Does

The **Portable Audit eXporter (PAX)** is an enterprise-grade PowerShell script that exports Microsoft Purview Unified Audit Log events, with specialized support for Microsoft 365 Copilot activities and related operations. It transforms raw audit data into analysis-ready CSV files with enriched metadata, intelligent query optimization, and flexible schema options.

Core Capabilities:

- Retrieves audit events from Microsoft 365 Unified Audit Log via Exchange Online Management
- Exports to structured CSV with optional array explosion and deep JSON flattening
- Includes enriched usage & ROI fields (tokens, models, latency, acceptance metrics)
- Supports both live querying and offline replay/transformation of previously exported data
- Implements adaptive time slicing to navigate service limits intelligently

- Provides detailed logging of all operations, warnings, and performance metrics
- Automatically handles ExchangeOnlineManagement module installation and connection

Execution Modes:

1. **Standard Mode** - One row per audit record (raw JSON preserved in **CopilotEventData** column)
2. **Array Explosion Mode** (**-ExplodeArrays**) - Canonical Purview 35-column schema with array elements expanded
3. **Deep Flatten Mode** (**-ExplodeDeep**) - 35-column base schema + fully flattened **CopilotEventData.*** columns
4. **Offline Replay Mode** (**-RAWInputCSV**) - Re-process previously exported raw audit CSV files without querying the service

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Key Features

Intelligent Query Management

- **Adaptive Block Sizing:** Automatically adjusts time window sizes based on data density
- **10K Limit Detection:** Identifies when Microsoft 365 service cap is reached and recommends mitigation
- **Automatic Subdivision:** Binary/aggressive splitting of dense time periods to maximize completeness
- **Throttle Resilience:** Exponential backoff with jitter for retry operations
- **Volume Classification:** Smart batching based on activity type (High/Medium/Low volume)

Data Processing & Output

- **Purview Schema Compliance:** Matches Microsoft Purview's canonical exploded schema structure
- **Deep JSON Flattening:** Optional recursive flattening of nested **CopilotEventData** structures
- **Agent Filtering:** Filter records by specific AgentId values or any agent-related activity
- **Streaming Export:** Memory-efficient chunked CSV writing for large datasets
- **UTF-8 Encoding:** Consistent UTF-8 (no BOM) output for cross-platform compatibility
- **Header Stability:** Always writes CSV header even when zero records match (ensures schema consistency)

Performance Optimization

- **Parallel Execution (PS7+):** Concurrent processing of multiple activity types with controlled throttling
- **Learned Block Sizes:** Per-activity and global adaptive sizing based on observed densities
- **Fast CSV Writer:** Direct **StreamWriter** usage eliminates repeated **Export-Csv** overhead
- **Schema Sampling:** Configurable initial sampling to optimize column discovery vs. memory usage

Operational Excellence

- **Composite Progress:** Single weighted percentage across Query/Explosion/Export phases
- **Detailed Logging:** Comprehensive log file with parameters, decisions, warnings, and metrics
- **Automated Setup:** Detects, installs, and connects ExchangeOnlineManagement module automatically
- **Offline Replay:** Transform previously exported raw CSVs without Exchange Online connection

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Use Cases

Adoption & Usage Analytics

- Track Microsoft 365 Copilot adoption across your organization
- Measure user engagement with AI features (interactions, token consumption, model usage)
- Identify power users and underutilized licenses
- Calculate ROI metrics based on time saved and acceptance rates

Compliance & Governance

- Audit Copilot interactions for regulatory compliance requirements
- Monitor data access patterns and sensitivity indicators
- Track plugin usage and custom GPT deployment
- Generate audit trails for security reviews
- Filter and analyze specific Copilot Studio declarative agent activity

Performance & Capacity Planning

- Analyze response latencies and model performance
- Identify throttling patterns and peak usage periods
- Forecast infrastructure needs based on token consumption trends
- Optimize block sizing for your tenant's data density

Data Integration & BI

- Export enriched data to Power BI, Azure Synapse, or data warehouses
- Join audit data with licensing information for coverage analysis
- Build custom dashboards with wide-schema flattened data
- Maintain historical archives with consistent schema over time

Development & Testing

- Offline replay mode for reproducible transformations
- Test schema changes against synthetic or sanitized datasets
- Validate data pipelines without querying production audit logs
- Develop downstream analytics without live tenant access

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Prerequisites

Requirement	Details	Notes
PowerShell	5.1 or 7+	7+ strongly recommended for parallel execution and UTF-8

Requirement	Details	Notes
Unified Audit Logging	Enabled in tenant	Verify in Microsoft Purview compliance portal
Permissions	View-Only Audit Logs or Audit Logs role	Least privilege: Use read-only audit role
Network Access	Microsoft 365 endpoints	Ensure firewall allows connections to *.protection.outlook.com
Execution Policy	Bypass or RemoteSigned	See Authentication Methods

Note: The script automatically handles ExchangeOnlineManagement module detection, installation, and connection. No manual setup required.

Permission Details

Minimum RBAC Requirements:

- **View-Only Audit Logs** role (read-only, recommended for production)
- **Audit Logs** role (if write operations needed elsewhere)
- Member of appropriate role groups in Microsoft Purview compliance portal

Why PowerShell 7+?

Feature	PowerShell 5.1	PowerShell 7+
Parallel Execution	✗ Not Available	☑ <code>ForEach-Object -Parallel</code>
UTF-8 Default	✗ Requires explicit encoding	☑ Native UTF-8
Performance	Baseline	🚀 30-50% faster JSON/pipeline
TLS/Cipher Support	Legacy protocols	☑ Modern TLS 1.3
Cross-Platform	✗ Windows only	☑ Windows/macOS/Linux
Side-by-Side Install	N/A	☑ Does not replace PS 5.1

Download PowerShell 7: <https://aka.ms/powershell>

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Installation & Setup

Download the Script

Download `PAX_Purview_Audit_Log_Processor_v1.6.0.ps1` and save to a working directory (e.g., `C:\Scripts\PAX\`).

First Run (Quick Start)

```
# PowerShell 7+ (recommended)
pwsh -File .\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -StartDate 2025-10-01 -
EndDate 2025-10-02

# Windows PowerShell 5.1
powershell -ExecutionPolicy Bypass -File
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -StartDate 2025-10-01 -EndDate 2025-
10-02
```

What Happens:

1. Script detects and installs ExchangeOnlineManagement module if needed
2. Interactive browser sign-in prompt (unless `-Auth` specified)
3. Queries Unified Audit Log for the specified date range
4. Exports to `C:\Temp\CopilotInteraction_<timestamp>.csv` (default path)
5. Creates matching `.log` file with detailed execution metrics

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Parameters Reference

Date & Time Parameters

`-StartDate` (string)

Purpose: UTC start date (inclusive) for audit log query or replay filter

Format: `yyyy-MM-dd` (e.g., `2025-10-01`)

Default (Live Mode): Previous full UTC day if both dates omitted

Default (Replay Mode): No filter applied if omitted

Example: `-StartDate 2025-10-01`

`-EndDate` (string)

Purpose: UTC end date (exclusive) for audit log query or replay filter

Format: `yyyy-MM-dd` (e.g., `2025-10-02`)

Default (Live Mode): Previous full UTC day + 1 if both dates omitted

Default (Replay Mode): No filter applied if omitted

Example: `-EndDate 2025-10-02`

Date Behavior:

- **Live mode:** If both dates omitted, defaults to previous full UTC day (midnight to midnight)
- **Live mode:** Must specify both or neither (partial specification rejected)
- **Replay mode:** Both dates optional; act as filters on `CreationDate` column
- **Time zone:** All dates interpreted as UTC; convert local times before invocation

Output & File Parameters

-OutputFile (string)

Purpose: Target CSV file path for exported audit data

Default: `C:\Temp\CopilotInteraction_<timestamp>.csv`

Use When: Specifying custom output location or naming convention

Example: `-OutputFile "C:\AuditData\Copilot_$(Get-Date -Format 'yyyyMMdd').csv"`

Notes: Parent directories created automatically if missing

Authentication Parameters

-Auth (string)

Purpose: Authentication method for Exchange Online connection

Valid Values: `WebLogin`, `DeviceCode`, `Credential`, `Silent`

Default: `WebLogin`

Use When: Automating scripts, using headless terminals, or SSO scenarios

Examples:

- `-Auth WebLogin` - Interactive browser sign-in (default)
- `-Auth DeviceCode` - Device code flow for headless/remote sessions
- `-Auth Credential` - Prompt for username/password (stored in memory only)
- `-Auth Silent` - Attempt cached token (fails if no valid token)

Not applicable in replay mode (authentication skipped when using `-RAWInputCSV`)

Query Behavior Parameters

-BlockHours (double)

Purpose: Initial time window size (hours) for each audit query chunk

Range: `0.016667` to `24.0` (1 minute to 24 hours)

Default: `0.5` (30 minutes)

Use When:

- Frequently hitting 10K limit (reduce to 0.25 or lower)
- Sparse historical data (increase to 2-8 hours for faster processing)
- Fine-tuning for tenant-specific data density

Examples:

- `-BlockHours 0.25` - Dense periods, many records
- `-BlockHours 4.0` - Sparse backfills, low activity

Notes: Script learns optimal sizes during execution; this is just the starting point

-ResultSize (int)

Purpose: Target number of records to retrieve per activity per time window

Range: 1 to 10000

Default: 10000

Use When:

- Managing memory usage (lower values = smaller batches)
- Testing with small samples
- Avoiding service throttling (reduce to 2500-5000)

Example: `-ResultSize 5000`

Notes: Actual results may be less; this is the requested maximum

-PacingMs (int)

Purpose: Delay (milliseconds) between paginated API calls

Range: 0 to 10000

Default: 0 (no artificial delay)

Use When:

- Experiencing frequent throttling errors
- Running during peak tenant usage
- Spreading load over time for politeness

Examples:

- `-PacingMs 250` - Moderate pacing
- `-PacingMs 500` - Conservative pacing

Notes: Increases total execution time proportionally

-ActivityTypes (string[])

Purpose: Array of audit log operation names to retrieve

Default: `@('CopilotInteraction')`

Use When:

- Querying multiple activity types in one run
- Analyzing cross-functional behaviors (Teams + Copilot)
- Comparative analysis across services

Examples:

- Single: `-ActivityTypes CopilotInteraction`
- Multiple: `-ActivityTypes CopilotInteraction,MessageSent,FileAccessed`
- Custom: `-ActivityTypes @('MeetingDetail','SearchQueryPerformed')`

Notes: See [Activity Types Reference](#) for common operations

-AgentId (string[])

Purpose: Filter audit records to include only those with specific AgentId value(s)

Default: Not set (no agent filtering)

Use When:

- Analyzing specific Copilot Studio declarative agents
- Tracking usage of particular agent implementations
- Filtering for known AgentId patterns

Examples:

- Single: `-AgentId "CopilotStudio.Declarative.T_4e671777-fa6c-601a-b416-df08b6ae4c14.03dc0b8b-a75a-4b77-86d7-98185a176d1b"`
- Multiple: `-AgentId "SYSTEM_CreateGPT.declarativeCopilot","CopilotStudio.Declarative.T_..."`

Notes:

- Works in both live query and replay modes
 - AgentId is a top-level field in AuditData JSON
 - Takes precedence if both `-AgentId` and `-AgentOnly` are specified
-

`-AgentOnly` (switch)

Purpose: Filter audit records to include only those with any AgentId present

Default: Off (no agent filtering)

Use When:

- Analyzing all agent-related activity regardless of specific agent
- Identifying records that involve Copilot Studio agents
- Filtering out non-agent Copilot interactions

Example: `-AgentOnly`

Notes:

- Works in both live query and replay modes
 - More inclusive than `-AgentId` (includes any record with AgentId field populated)
 - Combined with `-ActivityTypes` for refined filtering
-

Data Processing Parameters

`-ExplodeArrays` (switch)

Purpose: Enable Purview canonical 35-column exploded schema (array elements → rows)

Default: Off (standard 1:1 row mode)

Use When:

- Need one row per array element for pivoting
- Matching Microsoft Purview export format

- Preparing for relational BI tools

Example: `-ExplodeArrays`

Notes: Forced ON automatically in replay mode

-ExplodeDeep (switch)

Purpose: Enable deep flattening (35-column base + all `CopilotEventData.*` columns)

Default: Off

Use When:

- Maximum data extraction for BI/ML pipelines
- Need every nested field as a separate column
- Building wide-schema data warehouses

Example: `-ExplodeDeep`

Notes: Significantly increases CSV width; test with short date range first

Offline Replay Parameters

-RAWInputCSV (string)

Purpose: Path to previously exported raw Purview audit CSV for offline transformation

Default: Not set (live query mode)

Use When:

- Re-processing raw exports with different explosion settings
- Development/testing without live tenant access
- Reproducible transformations for auditing

Example: `-RAWInputCSV "C:\PreviousExports\Copilot_RAW_20251001.csv"`

Restrictions: Cannot combine with live query parameters (`-Auth`, `-BlockHours`, `-ResultSize`, `-PacingMs`, `-ParallelMode`, `-MaxConcurrency`, `-MaxParallelGroups`)

Allowed with RAWInputCSV: `-StartDate`, `-EndDate`, `-ActivityTypes`, `-AgentId`, `-AgentOnly`, `-OutputFile`, `-ExplodeDeep`, `-ExportProgressInterval`, `-StreamingSchemaSample`, `-StreamingChunkSize`

Parallel Execution Parameters (PowerShell 7+ only)

-ParallelMode (string)

Purpose: Control parallel execution of multiple activity types

Valid Values: `Off`, `On`, `Auto`

Default: `Off`

Use When:

- Processing multiple high-volume activity types

- Maximizing throughput on multi-core systems
- Need **Auto** heuristic to decide based on activity count

Examples:

- **-ParallelMode Auto** - Let script decide based on activity count and volume
- **-ParallelMode On** - Force parallel execution
- **-ParallelMode Off** - Sequential processing (PS 5.1 compatible)

-MaxConcurrency (int)

Purpose: Maximum concurrent threads per activity group

Range: 1 to 10

Default: 2

Use When: Fine-tuning parallel execution to avoid throttling

Example: **-MaxConcurrency 4**

-MaxParallelGroups (int)

Purpose: Maximum number of activity groups to process concurrently

Range: 1 to 5

Default: 3

Use When: Limiting total concurrent operations

Example: **-MaxParallelGroups 2**

Advanced Tuning Parameters

-StreamingSchemaSample (int)

Purpose: Number of initial records to sample for schema discovery

Range: 100 to 10000

Default: 2000

Use When:

- Wide schemas need more samples to discover all columns
- Narrow schemas can use smaller samples for faster processing

Example: **-StreamingSchemaSample 5000**

-StreamingChunkSize (int)

Purpose: Number of records to write per CSV flush operation

Range: 100 to 20000

Default: 5000

Use When:

- Managing memory usage (lower = more frequent flushes)
- Optimizing write performance (higher = fewer I/O operations)

Example: `-StreamingChunkSize 10000`

-ExportProgressInterval (int)

Purpose: Row interval for export progress updates

Range: 1 to 10000

Default: 10

Use When: Need more granular progress updates

Example: `-ExportProgressInterval 5`

Helper Parameters

-Help (switch)

Purpose: Display built-in help documentation

Example: `.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -Help`

Use When: Quick reference without opening documentation

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Authentication Methods

The script supports four authentication methods for Exchange Online:

1. WebLogin (Default)

Interactive browser-based authentication. Best for ad-hoc queries and interactive sessions.

```
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -Auth WebLogin -StartDate 2025-10-01  
-EndDate 2025-10-02
```

2. DeviceCode

Device code flow for headless/remote sessions or terminals without browser access.

```
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -Auth DeviceCode -StartDate 2025-10-  
01 -EndDate 2025-10-02
```

3. Credential

Username/password prompt. Credentials stored in memory only during script execution.

```
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -Auth Credential -StartDate 2025-10-  
01 -EndDate 2025-10-02
```

4. Silent

Attempts to use cached authentication token. Falls back to WebLogin if no valid token exists.

```
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -Auth Silent -StartDate 2025-10-01 -  
EndDate 2025-10-02
```

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Usage Examples

Basic Queries

```
# Standard mode - previous day (auto-default)  
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1  
  
# Specific date range  
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -StartDate 2025-10-01 -EndDate 2025-  
10-02  
  
# Custom output path  
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -StartDate 2025-10-01 -EndDate 2025-  
10-02 -OutputFile "C:\AuditData\Copilot.csv"  
  
# Multiple activity types  
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -StartDate 2025-10-01 -EndDate 2025-  
10-02 -ActivityTypes CopilotInteraction,MessageSent,FileAccessed
```

Exploded Schema Queries

```
# Array explosion (35-column Purview schema)  
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -ExplodeArrays -StartDate 2025-10-01  
-EndDate 2025-10-02  
  
# Deep flatten (maximum column extraction)  
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -ExplodeDeep -StartDate 2025-10-01 -  
EndDate 2025-10-02
```

Performance Tuning

```
# Reduce block size for dense data (hitting 10K limit)  
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -BlockHours 0.25 -StartDate 2025-10-  
01 -EndDate 2025-10-01
```

```
# Increase block size for sparse historical data
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -BlockHours 4.0 -StartDate 2025-09-01
-EndDate 2025-09-07

# Add pacing to reduce throttling
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -PacingMs 250 -StartDate 2025-10-01 -
EndDate 2025-10-02
```

Parallel Execution (PowerShell 7+ only)

```
# Auto-detect parallel benefit
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -ParallelMode Auto -ActivityTypes
CopilotInteraction,MessageSent,FileAccessed

# Force parallel with custom concurrency
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -ParallelMode On -MaxConcurrency 4 -
MaxParallelGroups 2 -ActivityTypes CopilotInteraction,MessageSent,FileAccessed
```

Offline Replay

```
# Basic replay (forced explosion)
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -RAWInputCSV
"C:\PreviousExports\Copilot_RAW.csv" -OutputFile
"C:\AuditData\Copilot_Exploded.csv"

# Replay with deep flatten and date filtering
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -RAWInputCSV
"C:\PreviousExports\Copilot_RAW.csv" -ExplodeDeep -StartDate 2025-10-01 -EndDate
2025-10-02 -OutputFile "C:\AuditData\Copilot_Deep.csv"

# Replay with activity filtering
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -RAWInputCSV
"C:\PreviousExports\Multi_Activity_RAW.csv" -ActivityTypes CopilotInteraction -
OutputFile "C:\AuditData\Copilot_Only.csv"

# Replay with agent filtering (any agent)
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -RAWInputCSV
"C:\PreviousExports\Copilot_RAW.csv" -AgentOnly -OutputFile
"C:\AuditData\Agent_Records.csv"

# Replay with specific agent ID
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -RAWInputCSV
"C:\PreviousExports\Copilot_RAW.csv" -AgentId
"CopilotStudio.Declarative.T_4e671777-fa6c-601a-b416-df08b6ae4c14.03dc0b8b-a75a-
4b77-86d7-98185a176d1b" -OutputFile "C:\AuditData\Specific_Agent.csv"
```

Agent Filtering (Live & Replay)

```
# Filter for any agent-related records (live query)
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -AgentOnly -StartDate 2025-10-01 -
EndDate 2025-10-02

# Filter for specific agent ID(s)
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -AgentId
"SYSTEM_CreateGPT.declarativeCopilot" -StartDate 2025-10-01 -EndDate 2025-10-02

# Multiple agent IDs with deep flatten
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -ExplodeDeep -AgentId
"SYSTEM_CreateGPT.declarativeCopilot","CopilotStudio.Declarative.T_..." -StartDate
2025-10-01 -EndDate 2025-10-02

# Agent filtering in replay mode
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -RAWInputCSV
"C:\PreviousExports\Copilot_RAW.csv" -AgentOnly -OutputFile
"C:\AuditData\All_Agents.csv"
```

Authentication Variations

```
# Device code for headless session
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -Auth DeviceCode -StartDate 2025-10-
01 -EndDate 2025-10-02

# Credential prompt
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -Auth Credential -StartDate 2025-10-
01 -EndDate 2025-10-02

# Silent (cached token)
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -Auth Silent -StartDate 2025-10-01 -
EndDate 2025-10-02
```

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Output Files & Schema

Output Files

Every execution produces two files:

1. CSV File (Data Export)

- **Location:** Specified by `-OutputFile` parameter (default: `C:\Temp\CopilotInteraction_<timestamp>.csv`)
- **Encoding:** UTF-8 without BOM
- **Format:** Standard CSV with quoted fields
- **Header:** Always written (even when zero records match)

- **Line Endings:** CRLF on Windows, LF on macOS/Linux

2. Log File (Execution Metrics)

- **Location:** Same directory as CSV, **.csv** replaced with **.log**
- **Contains:**
 - Script parameters and version
 - Authentication method and connection details
 - Query plan and adaptive block sizing decisions
 - Progress updates and phase transitions
 - Warnings (10K limits, throttling, schema changes)
 - Final metrics (records processed, time elapsed, throughput)

Schema Modes

Standard Mode (Default)

One row per audit record. CopilotEventData preserved as JSON string in a single column.

Column Count: Variable (base audit fields + **CopilotEventData** JSON column)

Use When: Need raw data for custom processing or minimal transformation

Exploded Mode (**-ExplodeArrays**)

Purview canonical 35-column schema. Array elements (Messages, AccessedResources, AISystemPlugins) expanded to separate rows.

Column Count: 35 base columns

Base Columns (35):

1. RecordId
2. CreationDate
3. RecordType
4. Operation
5. UserId
6. AssociatedAdminUnits
7. AssociatedAdminUnitsNames
8. AgentId
9. AgentName
10. AppIdentity_AppId
11. AppIdentity_DisplayName
12. AppIdentity_PublisherId
13. ApplicationName
14. CreationTime
15. ClientRegion
16. Audit_UserId
17. AppHost

18. ThreadId
19. Context_Id
20. Context_Type
21. Message_Id
22. Message_isPrompt
23. AccessedResource_Action
24. AccessedResource_PolicyDetails
25. AccessedResource_SiteUrl
26. AISystemPlugin_Id
27. AISystemPlugin_Name
28. ModelTransparencyDetails_ModelName
29. MessageIds
30. Message_AcceptanceState
31. Message_TokenCount
32. Message_CharacterCount
33. Message_LengthPreference
34. Message_LatencyMilliseconds
35. ExplosionTruncated

Use When: Need relational format for BI tools or matching Microsoft Purview exports

Deep Flatten Mode (-ExplodeDeep)

35 base columns + all nested CopilotEventData.* columns. Maximum data extraction with every nested field as a separate column.

Column Count: 35+ (dynamic based on data)

Use When:

- Maximum data extraction for BI/ML pipelines
- Need every nested field accessible as a column
- Building wide-schema data warehouses

Warning: Significantly increases CSV width and processing time. Test with short date range first.

Activity Types Reference

Common High-Volume Activities

- **CopilotInteraction** - Microsoft 365 Copilot usage events
- **MessageSent** - Teams/Exchange message sending
- **FileAccessed** - SharePoint/OneDrive file access
- **MailItemsAccessed** - Email access events

Common Medium-Volume Activities

- **MessageRead** - Message read receipts
- **FileModified** - File edit operations

- **MeetingDetail** - Teams meeting metadata
- **SearchQueryPerformed** - Search queries

Common Low-Volume Activities

- **CreatePlugin** - Copilot plugin creation
- **UpdatePlugin** - Plugin modifications
- **DeletePlugin** - Plugin removal
- **EnablePlugin** / **DisablePlugin** - Plugin state changes

Finding Available Activities

For a complete list of available Purview audit activities and operations, refer to the Microsoft Learn documentation:

 **[Audit log activities - Microsoft Purview](#)**

This comprehensive reference includes all available operations across Microsoft 365 services, including SharePoint, Exchange, Teams, Copilot, and more.

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Advanced Features

Adaptive Block Sizing

The script automatically adjusts time window sizes based on observed data density:

- **Initial Block:** Starts with **-BlockHours** parameter (default 30 minutes)
- **Learning Phase:** Monitors record counts per window
- **Automatic Subdivision:** Splits windows hitting the 10K service limit
- **Progressive Refinement:** Shrinks blocks for dense periods, expands for sparse periods
- **Per-Activity Learning:** Maintains separate learned block sizes for each activity type

Parallel Execution (PowerShell 7+)

When processing multiple activity types, parallel execution can significantly improve performance:

- **Auto Mode:** Script heuristically determines if parallel execution will benefit
- **Forced Mode:** Always use parallel execution regardless of activity count
- **Throttling Control:** Configurable concurrency limits to avoid overwhelming the service
- **Group Processing:** Activities classified by volume (High/Medium/Low) and processed in batches

Offline Replay Mode

Re-process previously exported raw audit CSV files without querying Exchange Online:

- **No Authentication Required:** Skip connection to Microsoft 365
- **Flexible Filtering:** Apply date, activity, and agent filters to existing data
- **Schema Transformation:** Convert raw exports to exploded or deep flatten schemas

- **Reproducible Analysis:** Test transformations against known datasets
- **Development Workflow:** Build pipelines without production access

Progress Tracking System

Real-time progress updates across three phases:

Display Format:

```
PAX Purview Audit Log Processing
Status: Query: 45/100(45%) | Explosion: 12000/25000(48%) | Export: 0/1(0%) :: 42%
```

Components:

- **Overall percentage:** Weighted composite across all phases
- **Phase detail:** Current/Total (percentage) for each active phase
- **Batch info:** Current batch number, estimated total, percentage range
- **Record range:** Shows which records currently processing (in batches)

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Performance Tuning

Hitting the 10K Service Limit

Symptoms:

- Log shows: `CRITICAL: 10K limit reached for time window <dates>`
- CSV may be incomplete for dense periods

Immediate Action:

```
# Reduce block hours to 15 minutes or less
pwsh -File .\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 `
  -BlockHours 0.25 `
  -StartDate 2025-10-03 `
  -EndDate 2025-10-03
```

Progressive Tuning:

1. Start: `-BlockHours 0.5` (30 min) → If still hitting: `0.25` (15 min)
2. If still saturated: `0.133333` (8 min) → `0.066667` (4 min)
3. Minimum: `0.016667` (1 min)

Verification:

- Check log for "Data retrieval completed without hitting limits"
- Compare record counts across runs

- Monitor **Hit10KLimit** flag in metrics section

Throttling & Rate Limiting

Symptoms:

- Log shows: **WARNING: Throttling detected, backing off...**
- Frequent retry attempts
- Extended execution times

Solutions:

```
# Add inter-page pacing (250ms delay between API calls)
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -PacingMs 250 -StartDate 2025-10-01 -
EndDate 2025-10-02

# Reduce ResultSize to smaller batches
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -ResultSize 5000 -StartDate 2025-10-
01 -EndDate 2025-10-02

# Combine both approaches
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -ResultSize 5000 -PacingMs 250 -
StartDate 2025-10-01 -EndDate 2025-10-02
```

Memory Optimization

For Deep Flatten with Wide Schemas:

```
# Increase schema sample, reduce chunk size
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -ExplodeDeep `
-StreamingSchemaSample 5000 `
-StreamingChunkSize 2000 `
-StartDate 2025-10-01 `
-EndDate 2025-10-02
```

For Narrow Schemas (Faster Processing):

```
# Reduce schema sample, increase chunk size
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -ExplodeArrays `
-StreamingSchemaSample 1000 `
-StreamingChunkSize 10000 `
-StartDate 2025-10-01 `
-EndDate 2025-10-02
```

Parallel Execution Tuning

Conservative Approach (Avoid Throttling):

```
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -ParallelMode On `
  -MaxConcurrency 2 `
  -MaxParallelGroups 2 `
  -ActivityTypes CopilotInteraction,MessageSent,FileAccessed
```

Aggressive Approach (Maximum Throughput):

```
.\PAX_Purview_Audit_Log_Processor_v1.6.0.ps1 -ParallelMode On `
  -MaxConcurrency 4 `
  -MaxParallelGroups 3 `
  -ActivityTypes CopilotInteraction,MessageSent,FileAccessed
```

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Troubleshooting & FAQ

Common Issues:

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Frequently Asked Questions:

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-

Common Issues

Authentication Failures

Problem: "Unable to connect to Exchange Online"

Solutions:

- Verify you have View-Only Audit Logs or Audit Logs role assigned
- Check network connectivity to *.protection.outlook.com
- Try different auth method: -Auth DeviceCode for headless sessions
- Clear cached credentials: Restart PowerShell session

No Data Returned

Problem: CSV contains only header, no records

Solutions:

- Verify Unified Audit Logging is enabled in your tenant
- Check date range (dates are UTC, not local time)
- Confirm activity type spelling: -ActivityTypes CopilotInteraction (case-sensitive)
- Verify users have generated audit events in the date range
- Check audit log retention period (default 90 days)

10K Limit Warnings

Problem: Log shows "CRITICAL: 10K limit reached"

Solutions:

- Reduce -BlockHours parameter (try 0.25 or 0.133333)
- Run script multiple times with shorter date ranges
- Check adaptive subdivision is working (log should show automatic splits)
- Consider if data is genuinely dense (may need multiple runs)

Memory Issues

Problem: Script consumes excessive memory or crashes

Solutions:

- Reduce -StreamingChunkSize (try 2000 or 1000)
- Increase -StreamingSchemaSample to discover schema earlier (try 5000)
- Avoid -ExplodeDeep for initial runs (use -ExplodeArrays instead)
- Process shorter date ranges
- Close other applications to free memory

Throttling Errors

Problem: Frequent "Throttling detected" messages

Solutions:

- Add pacing: -PacingMs 250 or -PacingMs 500
- Reduce ResultSize: -ResultSize 5000
- Run during off-peak hours
- Disable parallel mode if enabled

- Consider if tenant is under heavy load
-

Frequently Asked Questions

Q: Does the script modify any data?

A: No. The script is read-only and only exports audit data. No modifications are made to audit logs or tenant configuration.

Q: What timezone are dates in?

A: All dates are interpreted as UTC. Output timestamps are also UTC in ISO 8601 format (`yyyy-MM-ddTHH:mm:ss.fffZ`).

Q: Can I filter by specific users or models at the source?

A: No, filtering happens after export. The Unified Audit Log API does not support user/model filtering. Export data and filter in post-processing or BI tools.

Q: How deep does the script flatten JSON?

A: Standard explode: 60 levels. Deep flatten: 120 levels. JSON serialization: 60 levels. These are constants in the script and can be adjusted if needed.

Q: Can I run this in an automated schedule?

A: Yes. Use `-Auth Silent` with cached credentials or `-Auth Credential` with saved credentials. Consider using Task Scheduler (Windows) or cron (macOS/Linux).

Q: What if I need older audit logs?

A: Audit retention depends on your tenant's licensing. E3/E5 licenses retain 90-365 days. Check Microsoft Purview compliance portal for your retention period.

Q: Does the script work on macOS/Linux?

A: Yes, with PowerShell 7+. Install PowerShell 7 and ExchangeOnlineManagement module. Authentication methods may vary (WebLogin, DeviceCode recommended).

Q: How do I handle very large date ranges?

A: Break into smaller chunks (weekly or monthly), run separately, then concatenate CSV files. Use `-OutputFile` to name by date range.

Q: Can I customize the output schema?

A: The 35-column base schema is fixed to match Purview standards. In `-ExplodeDeep` mode, additional columns are auto-discovered from nested data.

Q: What's the difference between `-ExplodeArrays` and `-ExplodeDeep`?

A: `-ExplodeArrays` creates 35 columns with array elements as separate rows. `-ExplodeDeep` adds all nested `CopilotEventData.*` fields as additional columns (wide schema).

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Known Limitations

Area	Limitation / Behavior	Mitigation / Guidance
Unified Audit 10K cap	Each <code>Search-UnifiedAuditLog</code> window tops at 10,000 records	Script auto-subdivides; if still saturated, re-run with smaller <code>-BlockHours</code> ($\leq 30m$)
Row explosion cap	Per original record explosion capped at 1,000 rows (<code>ExplosionTruncated</code> flag)	Investigate fan-out; consider narrower date, filter operations, or deep analysis separately
JSON / flatten depth	JSON serialization depth fixed at 60; deep flatten recursion capped at 120	Extremely deep structures beyond caps truncated; adjust constants if required
Memory usage	Streaming, chunked export by default	Tune with <code>-StreamingSchemaSample</code> / <code>-StreamingChunkSize</code> ; shard by date for extreme spans
Replay mode	Non-exploded mode disabled; always at least exploded schema	Use live mode if raw 1:1 row shape required
Parallel mode	Only helps multi-activity sets; single high-volume activity remains serial	Add more activity types or accept serial path
Time zones	Dates interpreted as UTC; <code>yyyy-MM-dd</code> must be UTC	Convert local times to UTC prior to invocation to avoid DST drift
Streaming export	Always on (chunked)	Adjust sample/chunk sizes for schema width & memory balance

Additional Notes

Streaming Export Behavior:

- Samples initial records (default 2000) to finalize column schema
- Writes header once, then processes rows in chunks (default 5000)
- Auto-adjusts chunk size based on column count ($> 250/500/750/1000$ columns $\rightarrow$ smaller chunks)
- Boosts chunk size for narrow schemas (≤ 60 columns $\rightarrow$ up to 15K)
- New columns discovered after schema freeze are ignored (warning emitted)

Fast CSV Writer:

- Uses in-process UTF-8 `StreamWriter` with manual escaping

- No repeated **Export-Csv** invocations
- Significantly faster for large exports (>300K rows)
- Transparent to user (no parameter required)

Timestamp Normalization:

- All timestamps output in UTC
- ISO 8601 format with millisecond precision: **yyyy-MM-ddTHH:mm:ss.fffZ**
- Eliminates locale ambiguity
- Simplifies downstream parsing

Parallel Replay Explosion (PS 7+ only):

- When replaying raw CSV with large record count
- Switches to controlled parallel explosion
- Dynamic throttle + batch resizing (targets ~0.8s–2.5s batches)
- Metrics emitted at completion

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Security & Compliance

Data Handling

- **Read-Only Operations:** Script never modifies audit logs or tenant configuration
- **Credential Security:** Authentication credentials never written to disk (memory only)
- **Audit Trail:** All operations logged with timestamps and parameters
- **No External Services:** No data transmitted to third-party services
- **Local Processing:** All data processing occurs on the execution machine

Permissions & Least Privilege

Recommended:

- Use **View-Only Audit Logs** role (read-only)
- Create dedicated service account for automated runs
- Limit access to output files (contain sensitive audit data)
- Review Microsoft Purview RBAC documentation

Not Required:

- No Global Administrator rights needed
- No tenant configuration changes
- No write access to audit logs

Compliance Considerations

- **Data Residency:** Export files remain on execution machine (control geography)
- **Retention:** Manage CSV retention per organizational policies
- **Encryption:** Consider encrypting output files for sensitive environments

- **Access Control:** Limit script execution to authorized personnel
- **Audit Review:** Regularly review log files for anomalies

Security Best Practices

1. **Execution Policy:** Use `Bypass` or `RemoteSigned` (avoid `Unrestricted`)
2. **Script Verification:** Validate script hash before execution
3. **Credential Management:** Avoid storing passwords in scripts (use `-Auth Silent` or `-Auth DeviceCode`)
4. **Network Security:** Ensure TLS 1.2+ for Exchange Online connections
5. **Output Protection:** Store CSV files in encrypted volumes or secure file shares
6. **Access Logging:** Enable filesystem auditing for output directories

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Additional Resources

Microsoft Documentation

- [Microsoft Purview Audit \(Premium\)](#) - Overview of audit capabilities
- [Audit log activities](#) - Complete list of auditable activities
- [Search the audit log](#) - Audit log search basics
- [Exchange Online PowerShell](#) - Exchange Online module documentation

Related Tools

- [Power BI](#) - Visualize exported audit data
- [Azure Synapse Analytics](#) - Data warehousing for large audit datasets
- [Microsoft Sentinel](#) - SIEM integration for audit logs

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